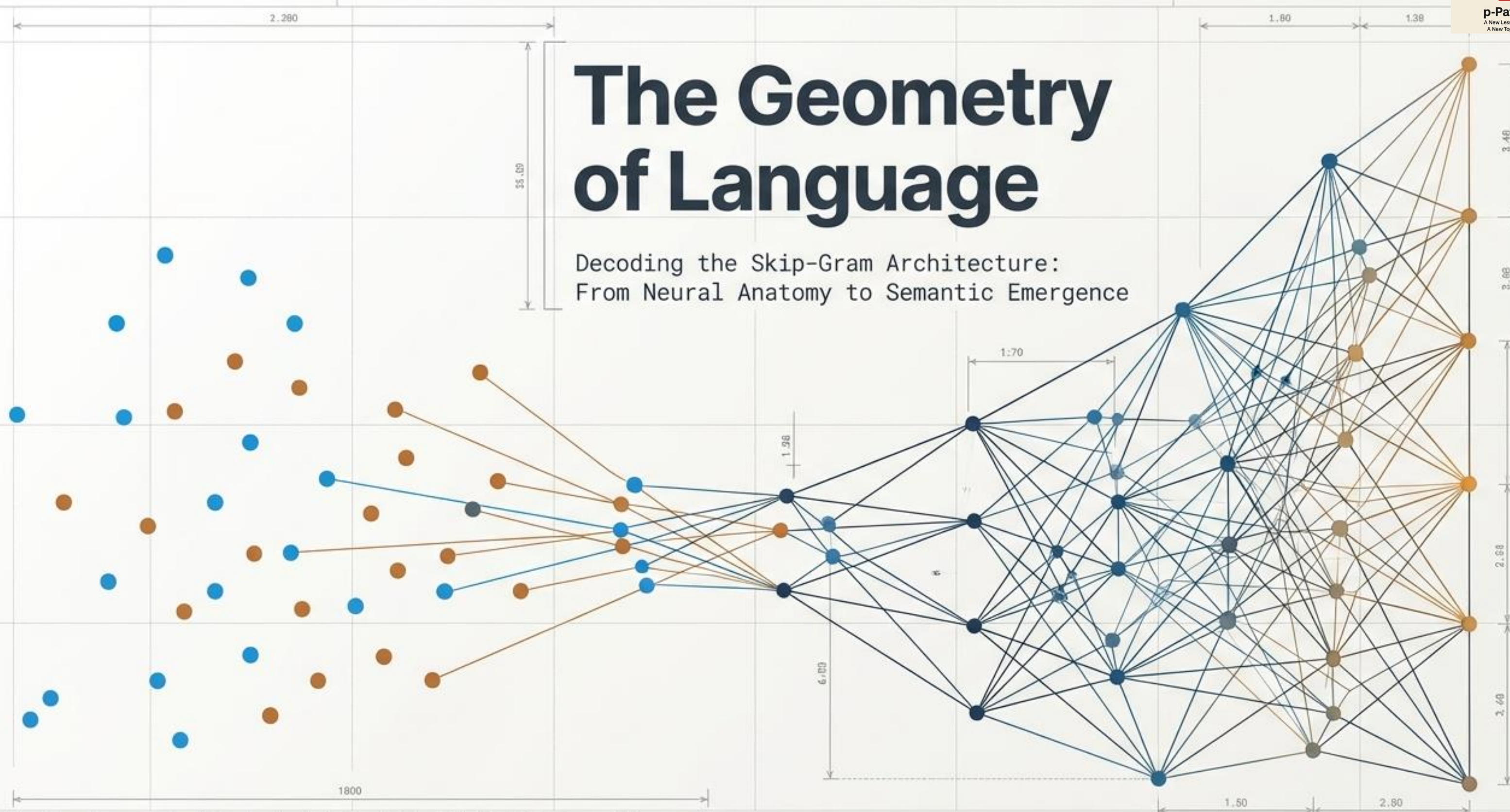


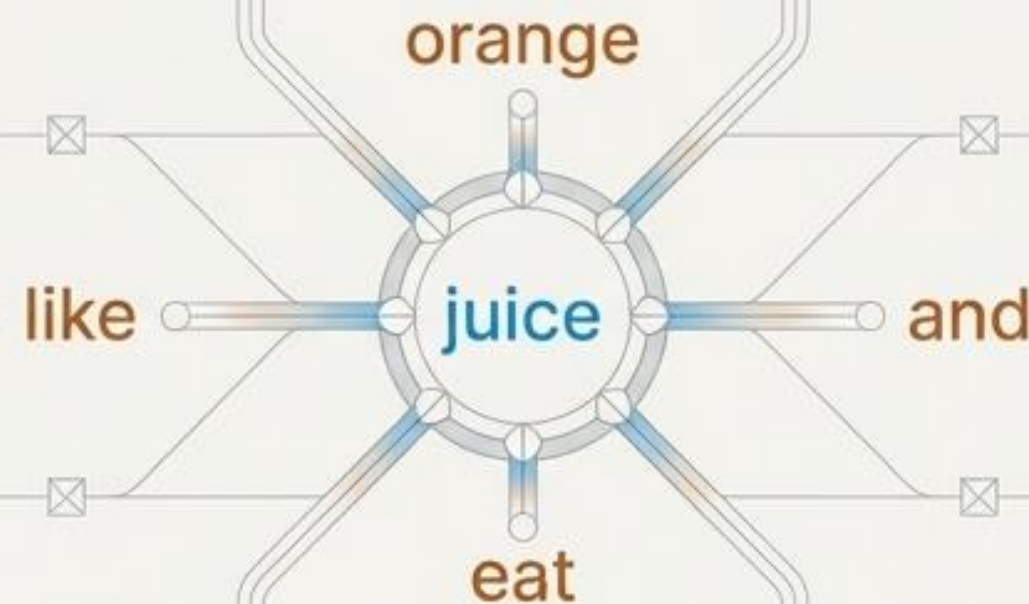


The Geometry of Language

Decoding the Skip-Gram Architecture:
From Neural Anatomy to Semantic Emergence



“You shall know a word by the company it keeps.”



The Intuition

Meaning does not reside in a word isolated by itself, but in the pattern of words that gather around it.

The Toy Sentence

I like orange juice and eat eggs daily.

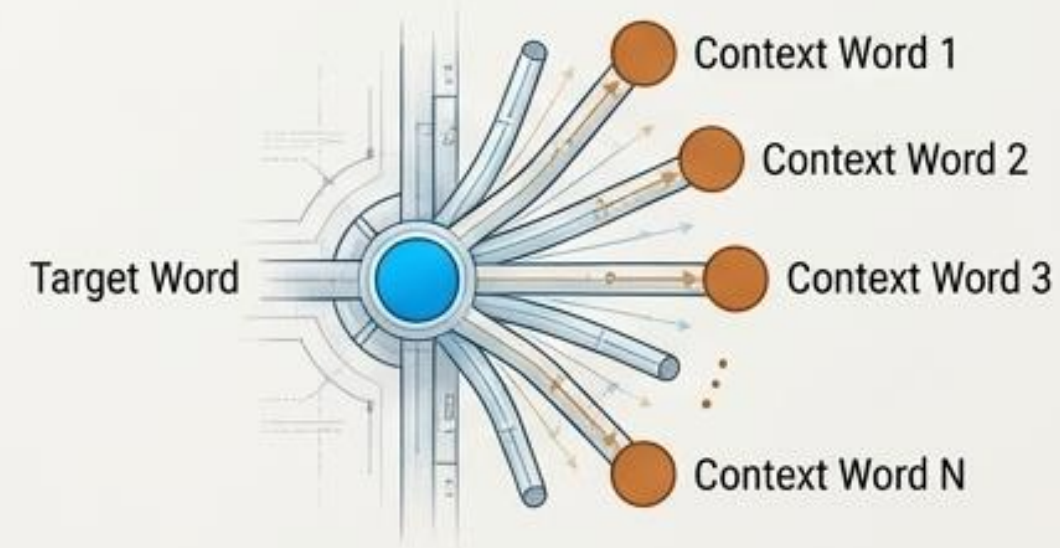
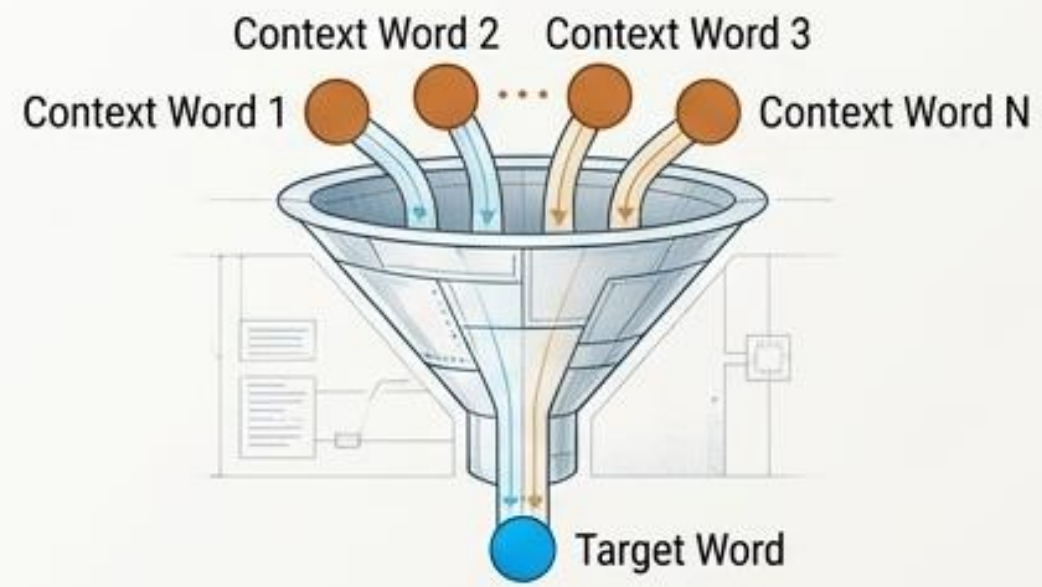
Target Word: juice

Context Window (Size 2): like, orange, and, eat

The Goal

Translate this linguistic co-occurrence into mathematical geometry.

Architectural Inversion: CBOW vs. Skip-Gram



Feature	CBOW (Continuous Bag-of-Words)	Skip-Gram
Objective	Predict target from context (Context → Target)	Predict context from target (Target → Context)
Input Processing	Averages multiple context words into one signal	Projects a single target word directly
Rare Word Handling	Poor (drowns out infrequent words during averaging)	Excellent (treats distinct target-context pairings as independent observations)
Computational Cost	Lower (faster training, fewer iterations)	Higher (generates multiple training samples from every single word)

The Power of Pairwise Extraction

The Delightful Problem

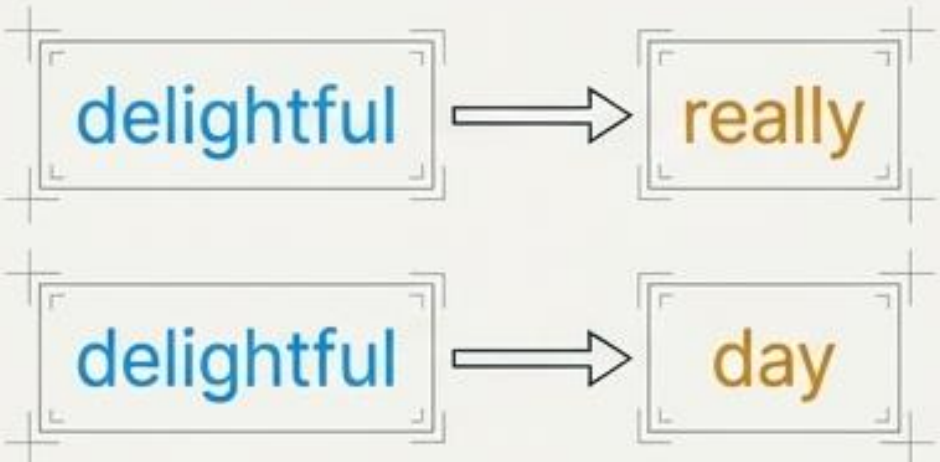
Corpus contains the phrase “a really delightful day.”

CBOW Approach



Averages context. The rare word “delightful” is ignored.

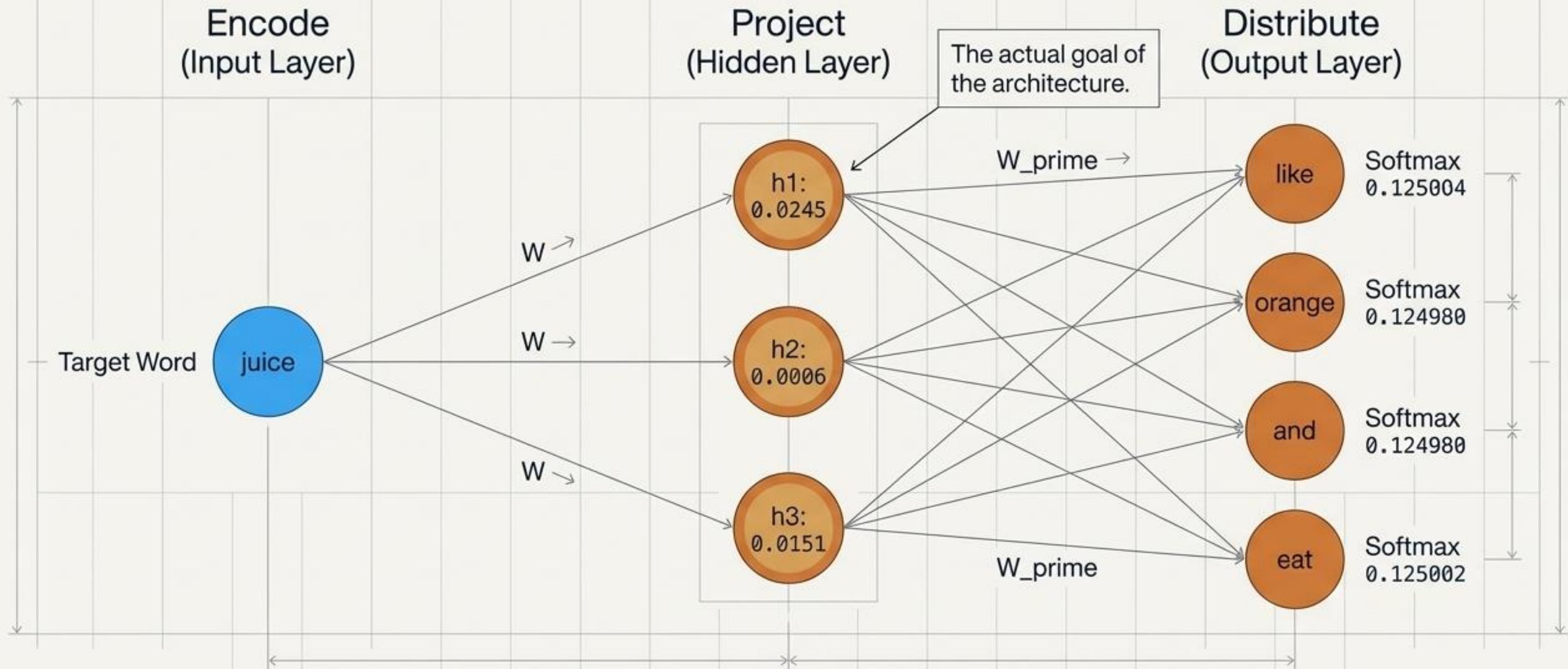
Skip-Gram Approach



Extracts independent pairs.

Because Skip-Gram does not force rare words to compete in an aggregated average, every word guarantees dedicated weight updates. It trades computational efficiency for localized, highly accurate representations.

Anatomy of a Forward Pass



Step 1: Encoding the Discrete Input

Sentence Vocabulary ($V=8$):

['I', 'like', 'orange', 'juice', 'and', 'eat', 'eggs', 'daily']

Target Word: **juice**

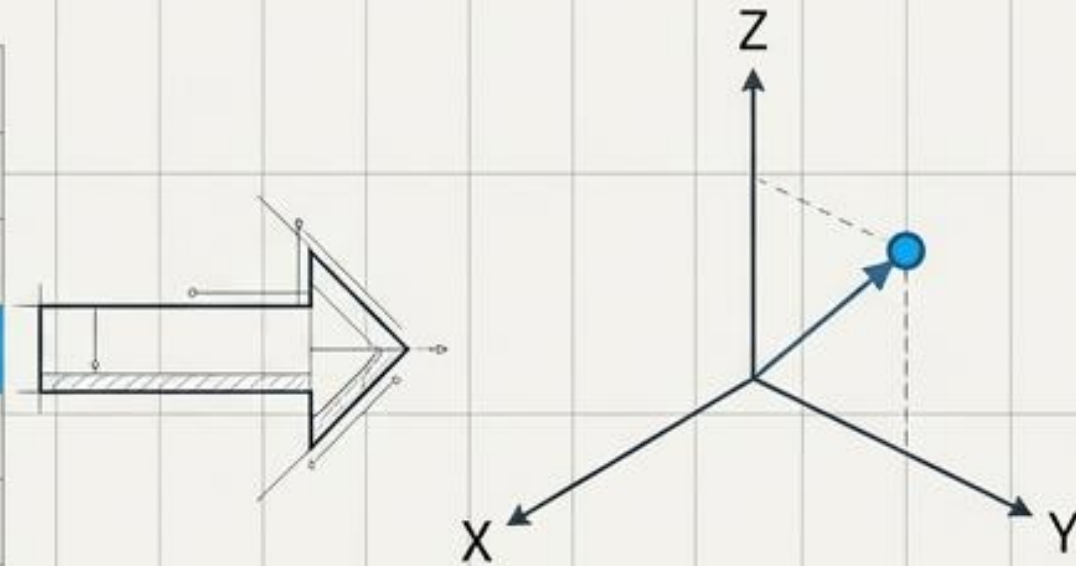
[0, 0, 0, **1**, 0, 0, 0, 0]

The network cannot read text. It requires a mathematical placeholder.
The $1 \times V$ One-Hot Vector is orthogonal to all other words—at this stage, the model assumes zero relationship between **juice** and any other word.

Step 2: The Hidden Space (The Real Objective)

$$\mathbf{x} \text{ (One-Hot)} \times \mathbf{W} \text{ (Weight Matrix)} = \mathbf{h} \text{ (Hidden Vector)}$$

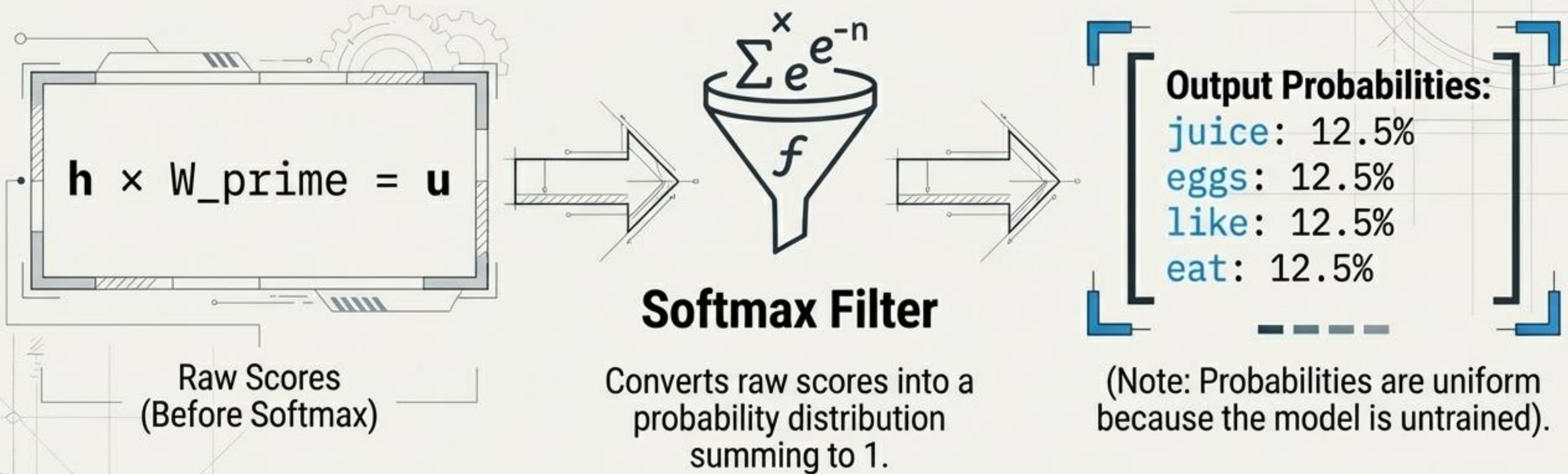
w11	w12	w13
w21	w22	w23
w31	w32	w33
0.0245	0.0006	0.0151
w51	w52	w53
w61	w62	w63
w71	w72	w73
w81	w82	w83



$$\mathbf{h} = [0.0245, 0.0006, 0.0151]$$

The final goal of Skip-Gram is not the output prediction. The prediction is just a training mechanism. Once training is complete, we discard the output layer and keep Matrix W. This matrix is the embedding space.

Step 3: Generating the Output Distribution



The Feedback Loop: The model compares this random distribution to the actual context ('like', 'orange', 'and', 'eat') to calculate the error.

The Proof of Learning: Meaning is a Distribution



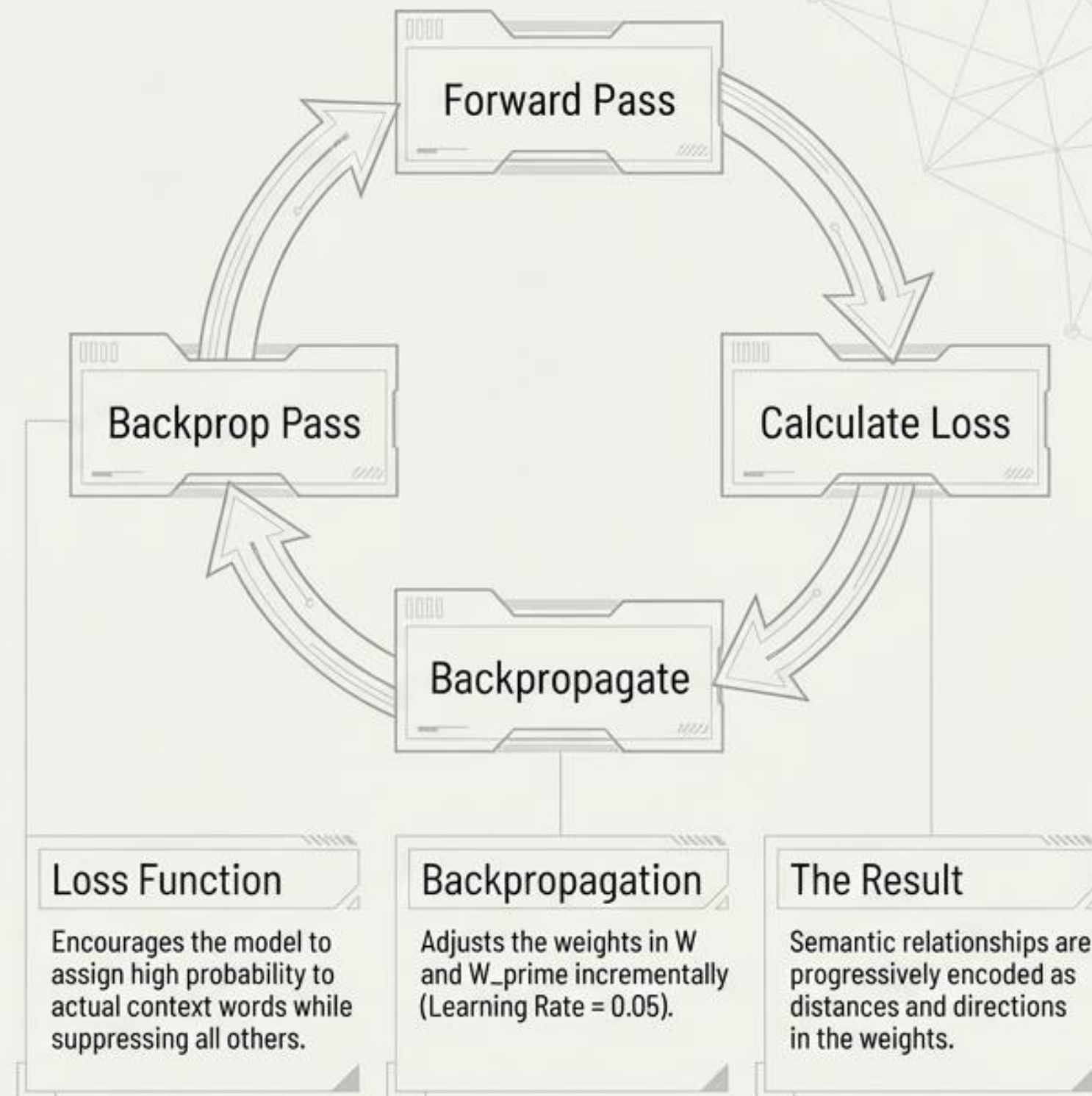
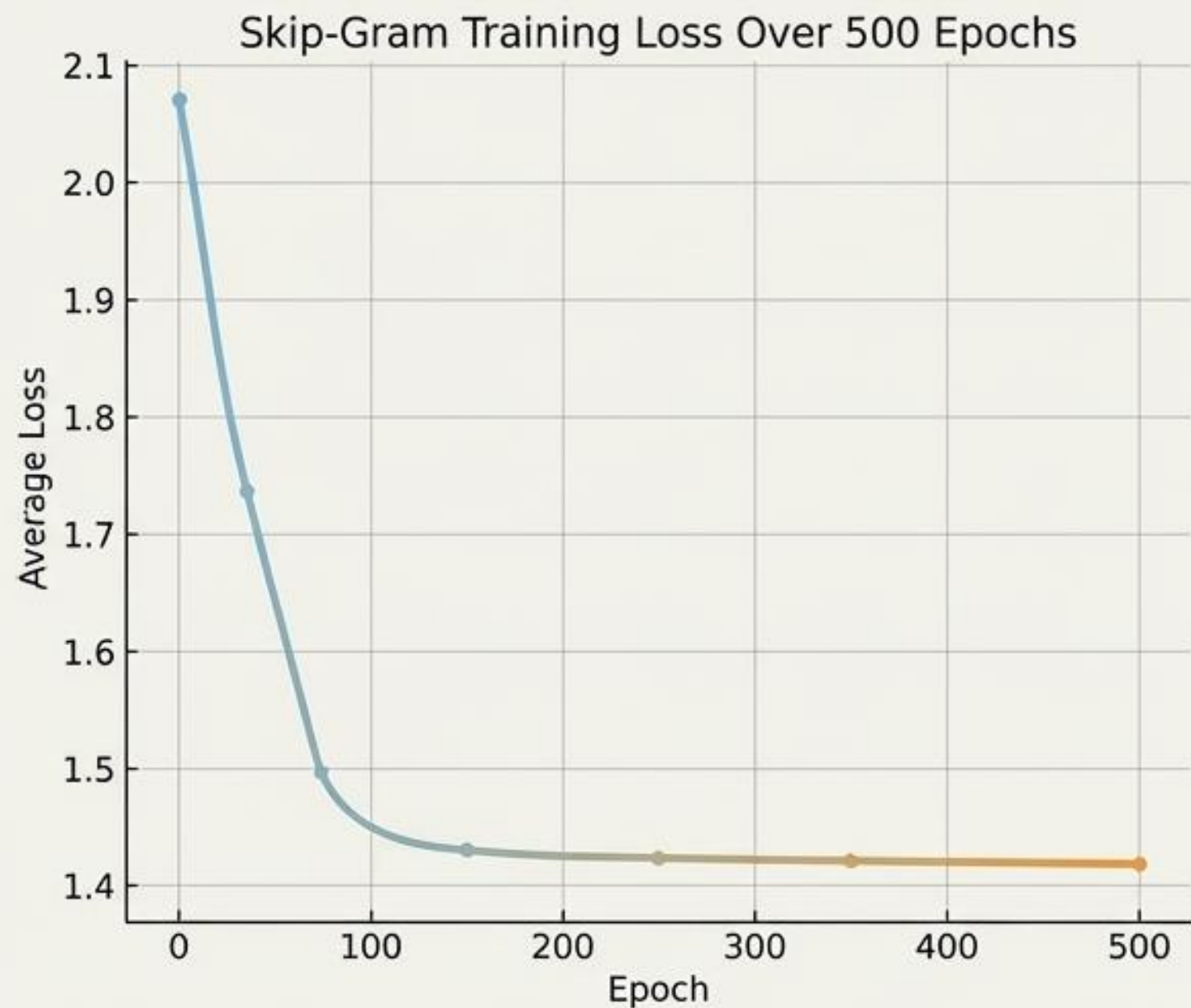
~50% random mass assigned to the true context words. (A flat, uniform distribution of noise).



>95% probability mass assigned directly to the true context words ('eat', 'orange', 'and', 'like').

Skip-Gram is best understood not as a model that predicts one neighboring word, but as a model that concentrates probability mass on the set of true neighbors. Meaning is not a prediction—it is a distribution.

The Training Cycle



The Nascent Geometry of the Toy Space



Contextual Alignment

'juice' and 'eggs' (0.68). Both appear as objects of consumption in the sentence.

Positional Proximity

'and' and 'daily' (0.84).

Contextual Dissonance

'like' and 'eggs' (-0.92). These words rarely share similar surrounding contexts in this training data.

Takeaway: Even from a single sentence, the embeddings have organized words into a meaningful vector space based entirely on distance and angle.

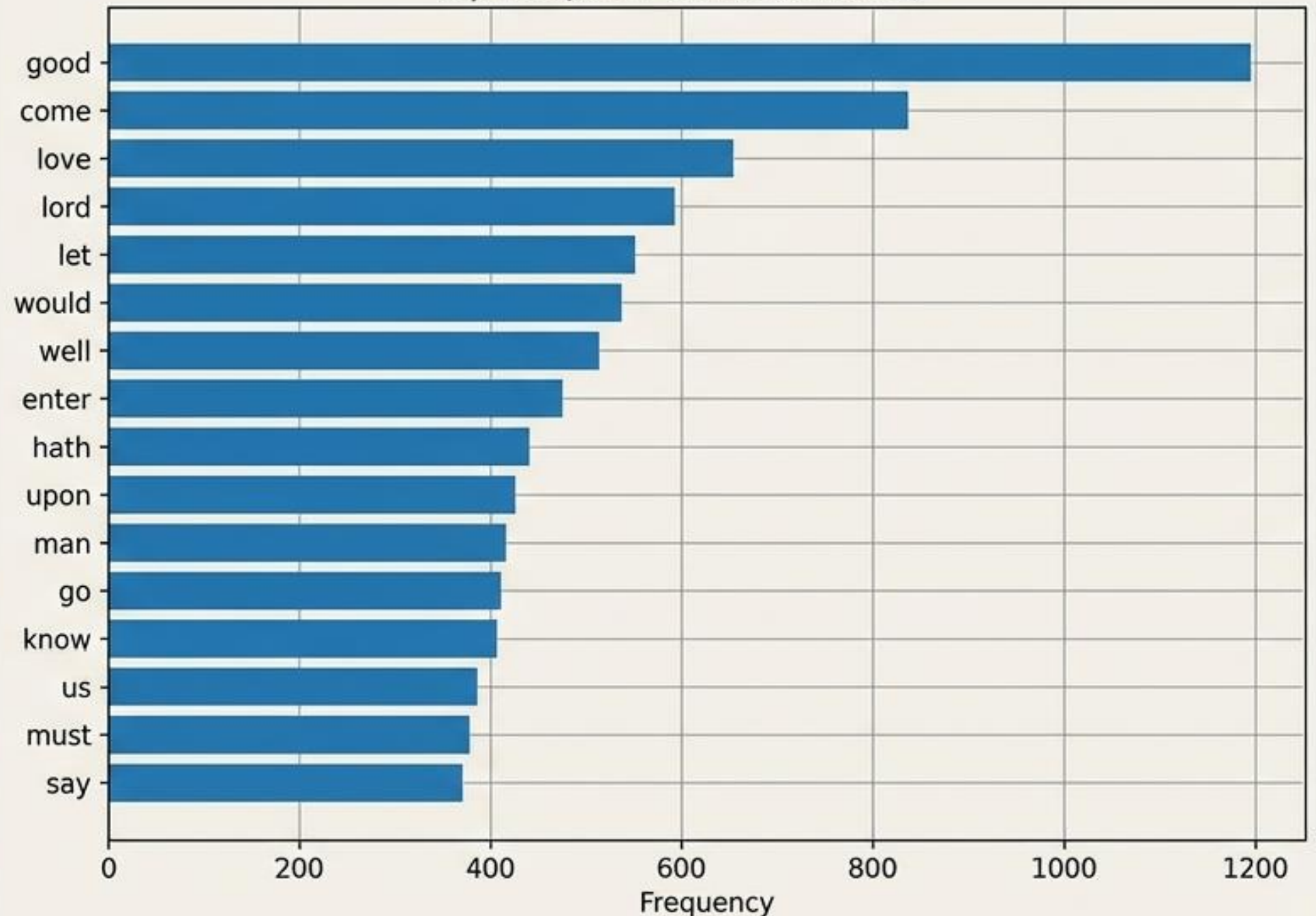
Scaling Up: The Shakespeare Corpus

Dataset & Architecture Stats

- Corpus Size:
8 Plays, 28,762 sentences
- Vocabulary Size:
2,998 unique words
- Embedding Dimensions (N):
100
- Context Window:
5

Moving from a synthetic example to authentic, complex linguistic patterns to uncover the true geometry of language.

Top Frequent Words (Filtered)



Emergent Associations in High-Dimensional Space

Target Word	Nearest Neighbors	Callout
trusted	game (0.977), knaves (0.977), defect (0.974), lawful (0.974)	Reflects contexts of judgment, loyalty, and moral evaluation.
virtue	merit (0.918), image (0.918), burst (0.918), ambition (0.916)	Aligns strongly with ethical and character-driven discourse.
brains	passage (0.946), changes (0.942), losses (0.939), stomach (0.938)	Captures metaphorical contexts of change, emotion, and human behavior, rather than anatomy.

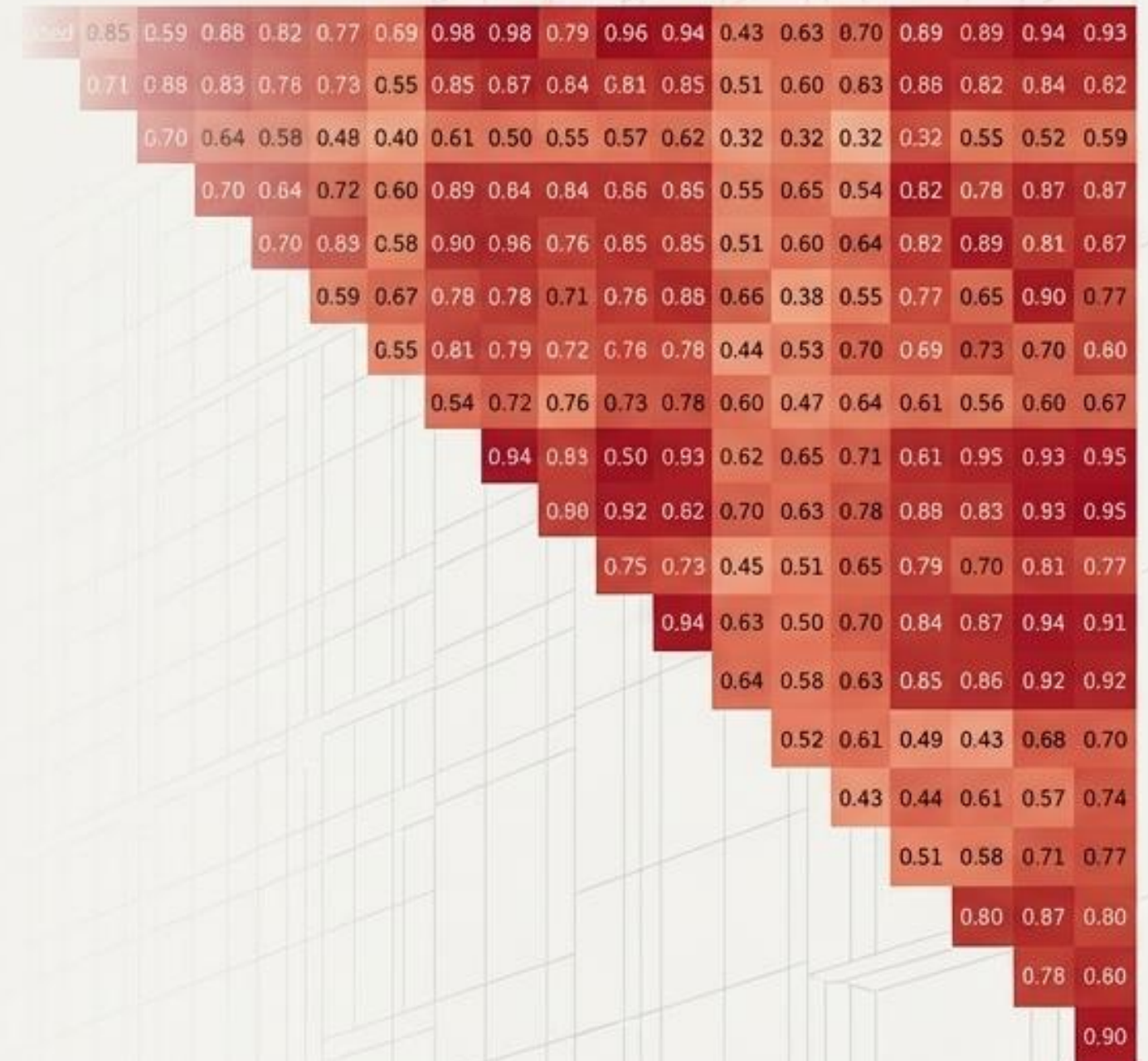
Takeaway: The model groups words based on shared narrative and functional contexts, independent of explicit programming.

Co-occurrence Transformed into Geometry

The similarity structure of language is not explicitly programmed; it emerges entirely from repeated exposure to contextual relationships.

Under the Skip-Gram objective, words that share linguistic environments are pulled together, while unrelated words are pushed apart.

Final Takeaway: Skip-Gram proves that meaning can be modeled as proximity in high-dimensional space. Embeddings are the definitive bridge between raw text and machine understanding.



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